

AWS Prerequisites Guide

America's Blood Centers AI Chatbot

Overview

This document outlines the AWS services and permissions required in your AWS account before deploying the America's Blood Centers AI Chatbot. The deployment will be automated through AWS CodeBuild, so no local tools are required on your end.

1. AWS Account Requirements

1.1 Active AWS Account

- Active AWS account with billing enabled
 - Access to or your preferred region
 - Account must not be in a restricted or suspended state
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2. Required AWS Services Access

The following AWS services must be available and accessible in your account:

2.1 Core AI/ML Services

- **Amazon Bedrock** - AI model hosting and knowledge base
- **Amazon Bedrock Knowledge Base** - Document and web content indexing
- **OpenSearch Serverless** - Vector database for semantic search

2.2 Compute Services

- **AWS Lambda** - Serverless functions for chat and sync operations

AWS CodeBuild - Automated deployment pipeline

2.3 Storage Services

- **Amazon S3** - Document storage and build artifacts
- **Amazon DynamoDB** - Conversation history storage

2.4 Frontend Hosting

- **AWS Amplify** - React frontend hosting and deployment

2.5 Security and Authentication

- **Amazon Cognito** - Admin user authentication
- **AWS IAM** - Identity and access management
- **AWS Secrets Manager** (optional) - Secure credential storage

2.6 Networking and Integration

- **Amazon EventBridge** - Scheduled daily synchronization
- **Amazon CloudWatch** - Logging and monitoring

2.7 Infrastructure Management

- **AWS CloudFormation** - Infrastructure deployment via CDK
 - **AWS Systems Manager (SSM)** - Parameter storage
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3. Amazon Bedrock Model Access (CRITICAL)

3.1 Required Models

You **MUST** request access to the following Bedrock models before deployment:

Model 1: Anthropic Claude Sonnet 4.5

- **Model ID:** `global.anthropic.claude-sonnet-4-5-20250929-v1:0`
- **Purpose:** Main conversational AI for chat responses
- **Required:** Yes

Model 2: Amazon Titan Embeddings Text v1

- **Model ID:** `amazon.titan-embed-text-v1`
- **Purpose:** Text embeddings for semantic search
- **Required:** Yes

3.2 How to Request Model Access

Method 1:

To get started, simply select a model from the Model catalog and open it in the playground or invoke the model using the [InvokeModel](#) or [Converse API](#) operations. Review [documentation](#) for the complete list of available models.

4. IAM Permissions Required

4.1 Deployment User/Role

The IAM user or role used for deployment must have permissions for the following services:

Required IAM Permissions:

- Amazon Bedrock (full access)
- OpenSearch Serverless (full access)
- AWS Lambda (full access)
- Amazon S3 (full access)
- Amazon DynamoDB (full access)
- Amazon Cognito (full access)
- AWS Amplify (full access)
- AWS CodeBuild (full access)
- Amazon EventBridge (full access)
- AWS CloudFormation (full access)
- AWS IAM (create/manage roles and policies)
- Amazon CloudWatch Logs (full access)
- AWS Systems Manager (parameter store access)

4.2 Recommended Approach

Attach **AdministratorAccess** policy to deployment user/role

5. Service Quotas and Limits

5.1 Verify Current Quotas

Please check and request increases if needed:

Service	Quota	Minimum Required	Recommended
Lambda	Concurrent executions	10	100
Lambda	Function storage	1 GB	5 GB
S3	Buckets	5	10
DynamoDB	Tables	2	5
Cognito	User pools	1	2
OpenSearch Serverless	Collections	1	2
OpenSearch Serverless	OCU (Compute Units)	2	4
CodeBuild	Concurrent builds	1	5

5.2 How to Check Quotas

1. Go to **Service Quotas** in AWS Console
2. Search for the service (e.g., "Lambda")
3. View current limits
4. Request increase if needed (click "Request quota increase")

7. Cost Estimation

7.1 Expected Monthly Costs

Development/Testing Environment:

- Estimated: \$380–\$420 per month

Production Environment (Low Traffic):

- Estimated: \$400–\$500 per month
- Assumptions: ~1,000 conversations/month

Production Environment (High Traffic):

- Estimated: **\$500–\$800 per month**
- Assumptions: ~10,000 conversations/month

7.2 Cost Breakdown by Service

Service	Estimated Monthly Cost
AWS Bedrock (Claude)	\$20 - \$200
OpenSearch Serverless	\$300 - \$400
AWS Lambda	\$5 - \$20
Amazon S3	\$1 - \$5
Amazon DynamoDB	\$5 - \$20
AWS Amplify	\$5 - \$15
Other Services	\$5 - \$10